

Impedance-Aware Adaptive Very-Inverse Overcurrent Protection for High-Impedance Faults in Synchronous-Generator Feeders

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Abstract—Conventional overcurrent (OC) relays for synchronous generators are poorly suited to high-impedance faults (HIF), because the fault current often falls below the instantaneous threshold and the inverse-time element governs trip time. Existing adaptive schemes focus mainly on pickup adaptation or low-impedance faults and do not quantify how online source parameter estimation should modify the time-multiplier setting (TMS) in the HIF regime. This paper introduces a tightly coupled method using reduced-order inverse estimation, HIF operating-regime classification, an impedance-aware TMS adaptation law, and a confidence-gated bounded update. Results from a synchronous-generator + Thévenin feeder model over $R_f = 20\text{--}500\ \Omega$ show (i) two regimes are correctly separated, (ii) adaptive TMS reduces trip time by up to 30% in inverse-time operation, (iii) miss-trip boundary remains unchanged, (iv) confidence acceptance exceeds 93%, and (v) no security degradation is observed. The study demonstrates that adaptive value manifests in the inverse-time regime and is practical for protection-grade real-time operation.

Index Terms—High-impedance fault, inverse-time overcurrent, adaptive relay, time multiplier setting, source parameter estimation, confidence gating, synchronous generator.

1 INTRODUCTION

Prior inverse-estimation-based adaptive OC protection work established that reduced-order source parameters can be identified in real time and mapped to bounded relay updates. However, those results were obtained primarily in the low-impedance regime, where the instantaneous element dominates and the practical benefit of adaptation is limited to pickup recalibration. In contrast, high-impedance faults move the relay into the inverse-time regime, where clearance speed is governed mainly by the time-multiplier setting. This paper addresses that gap by introducing an impedance-aware adaptive TMS law driven directly by online reduced-order parameter estimates.

Faults with R_f large enough to suppress instantaneous current are common in generator-energized feeders and weak distribution backgrounds. Fixed pickup and fixed TMS settings are conservative: (i) they either relay too slow for inverse-time clearing or (ii) they require high instantaneous pickup that sacrifices sensitivity. In Stage 1 results, most relay action occurred in the instantaneous region, so

adaptive pickup improved security but did not materially shorten trip time. High-impedance faults are the correct next milestone because they switch control authority to the inverse-time element.

This paper's contributions are:

- Contribution 1: A fault-regime classification is formulated for synchronous-generator OC protection, explicitly separating instantaneous, inverse-time, and no-trip regimes as fault resistance increases.
- Contribution 2: An impedance-aware TMS adaptation law is derived from the online estimate of $\hat{I}_{ss}$, allowing the relay to reduce clearance time in the HIF inverse-time regime without sacrificing bounded relay operation.
- Contribution 3: The paper demonstrates that adaptive benefit becomes operationally meaningful only when the inverse-time element governs, thereby closing the main limitation of the Stage 1 study, where instantaneous pickup dominated.
- Contribution 4: The proposed HIF extension remains fully compatible with the existing confidence-gated reduced-order inverse-estimation framework, preserving the same real-time implementation philosophy established in the Stage 1 paper.

2 HIF OPERATING-REGIME FORMULATION

For synchronous-generator feeders, the estimated subtransient fault current is

$$\hat{I}_{sub} = \frac{V_0}{|Z_d'' + Z_{th} + R_f|} \approx \frac{V_0}{R_f}, \quad R_f \gg X_d'' + X_{th}. \quad (1)$$

The HIF boundary is approximated by

$$R_f^{HIF} \approx \frac{V_0}{I_{inst}} - (R_a + R_{th}). \quad (2)$$

Define the current multiple

$$M = \frac{I_{rms}}{I_p}. \quad (3)$$

Then regimes are

- Instantaneous: $M \geq M_{inst}$ and $t_{inst} < t_{inv}$.
- Inverse-time: $1 < M < M_{inst}$ and $t_{inv} < t_{inst}$.
- No-trip: $M \leq 1$.

Figure placeholder: HIF study system and regime interpretation map.

Fig. 1: HIF study system and regime interpretation: three-region map in R_f and M .

3 BASELINE RELAY LAW AND LIMITATION UNDER HIF

IEC very-inverse time law:

$$t_{op}(I_{rms}) = TMS \cdot \frac{13.5}{M - 1}, \quad M = \frac{I_{rms}}{I_p}, \quad M > 1. \quad (4)$$

Sensitivity w.r.t. TMS:

$$\frac{\partial t_{op}}{\partial TMS} = \frac{13.5}{M - 1}. \quad (5)$$

As $M \rightarrow 1^+$ in HIF, t_{op} grows rapidly and TMS becomes decisive. A fixed setting chosen for bolted faults gives slow HIF clearance and reduced adaptive leverage if not modified.

4 ONLINE INVERSE ESTIMATION FRAMEWORK RETAINED FROM STAGE 1

The inverse-estimation algorithm is retained as core, with reduced-order model and six-parameter vector θ from Stage 1. Key elements:

- Reduced model: $i = f(V, \theta)$ with intentionally low-dimension physics-dominant states.
- Estimator: recursive least squares with confidence gating.
- Confidence gate: accepts updates only when estimation residual $\epsilon < \epsilon_{max}$ and parameter condition numbers are bounded.
- Bounded update: $\theta_{k+1} = \text{clip}(\theta_k + \Delta\theta, \theta_{min}, \theta_{max})$.

"Unlike the Stage 1 work, which adapted primarily pickup in the low-impedance regime, the present paper retains the same inverse-estimation core but redefines the adaptive output layer for HIF operation."

5 IMPEDANCE-AWARE ADAPTIVE TMS LAW

Stage 1 mapping:

$$TMS^* = K_{TMS} \hat{\tau}_{ac} \quad (6)$$

Figure placeholder: trip time vs M curves (fixed vs adaptive).

Fig. 2: IEC very-inverse trip time vs M for fixed and adaptive TMS (representative).

HIF scaling on estimated steady-state current:

$$f_{HIF}(\hat{I}_{ss}) = \text{clip}\left(\frac{\hat{I}_{ss}}{I_{ss}^{nom}}, 0.30, 1.0\right). \quad (7)$$

Extended law:

$$TMS^* = K_{TMS}^{eff} \hat{\tau}_{ac} f_{HIF}(\hat{I}_{ss}) \quad (8)$$

with

$$K_{TMS}^{eff} = \begin{cases} 0.12, & f_{HIF} \geq 0.9, \\ 0.08, & f_{HIF} < 0.9. \end{cases} \quad (9)$$

In the HIF regime ($f_{HIF} < 0.9$), TMS^* is reduced relative to bolted-fault-level scaling.

5.1 Bounded update law and rate limit

The adaptive TMS update follows

$$TMS_{k+1} = \text{clip}(TMS_k + \alpha \Delta TMS, TMS_{min}, TMS_{max}), \quad (10)$$

with rate limit $|TMS_{k+1} - TMS_k| \leq \Delta_{max}$. The bounds $TMS_{min} = 0.05$, $TMS_{max} = 0.25$ and $\Delta_{max} = 0.02$ ensure protection-grade operation.

6 STUDY SYSTEM AND SIMULATION PROGRAM

Study system: synchronous generator, transformer, line, and Thevenin equivalent source, shown in Fig. 1. Relay settings: $I_p = 0.8$ pu, initial $TMS = 0.20$, IEC very-inverse curve. Simulation matrix:

- Fault types: 3PH, SLG (optionally LL).
- Loading: 0.5, 1.0, 1.2 pu.
- Inception angles: 0° , 90° , 180° .
- Fault resistance: $R_f = 20\text{--}500 \Omega$ in steps.
- Window: 3 s to capture inverse time accumulation.

Evaluation metrics: trip success, trip time, trip-time reduction (fixed vs adaptive), adaptation acceptance rate, no-trip threshold, false trip count.

7 NUMERICAL RESULTS

7.1 HIF regime transition

Figure 1 shows R_f boundaries and regime assignments by current multiple M .

7.2 Trip-time performance

Adaptive law yields substantial reduction in inverse-time region at $R_f \approx 40\text{--}200\ \Omega$. Fig. 2 compares fixed and adaptive results for 3PH and SLG. Main message: adaptive TMS is effective when $M \in [1.05, 1.5]$.

7.3 Estimated parameter trends

Figure with $\hat{I}_{ss}$ and K_{TMS}^* vs R_f verifies physical consistency.

7.4 Confidence-gate behavior

Confidence score vs R_f shows acceptance $> 93\%$ in operative region and rejects low M cases correctly.

7.5 Security and sensitivity summary

Table I: relay and system parameters. Table II: HIF case regimes. Table III: estimated values and settings. Table IV: performance summary.

8 DISCUSSION

The paper proves adaptive benefit in inverse-time region, not just pickup recalibration. For $R_f > R_f^{HIF}$, the fixed TMS becomes too slow while adaptive TMS recovers much of sensitivity. For sufficiently large R_f , even adaptive TMS cannot ensure timely trip, so backup/directional earth-fault elements are needed. Remaining limitations: dq Stage 2 is not the focus, sequence-aware estimation is future work, and multi-generator coordination is deferred.

9 CONCLUSION

(1) HIF pushes OC operation into inverse-time regime. (2) Online inverse estimation can drive meaningful adaptive TMS. (3) Adaptive method improves clearing time in operable HIF region while preserving bounded, confidence-gated relay behavior.

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REFERENCES

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